

Question1. Explain how Simple Linear Regression works."

- **Simple Linear Regression is a supervised learning algorithm used to predict a continuous numerical target.**
- It uses **one independent variable (X)** to predict **one dependent variable (Y)**.
- It assumes that there is a **linear relationship between X and Y**.
- The relationship is represented using the equation:

$$\hat{Y} = \theta_0 + \theta_1 X$$

- **Theta0** is the intercept, which represents the predicted value of Y when X is zero.
- **Theta1** is the slope/coefficient, which tells us how much Y changes for a one-unit change in X.
- The algorithm finds the **best-fitting line** by minimizing the difference between actual and predicted values.
- These differences are called **residuals or errors**.
- It commonly uses the **Ordinary Least Squares (OLS)** approach to minimize the **sum of squared residuals**.
- Once the coefficients are learned, we can use the equation to **predict Y for new X values**.
- The model's performance can be evaluated using metrics such as **MAE, MSE, RMSE, and R2R^2**.

Simple Linear Regression – Formulas Students Should Remember

1. Mean of X

$$\bar{X} = \sum X / n$$

2. Mean of Y

$$\bar{Y} = \sum Y / n$$

3. Slope (b₁)

$$b_1 = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{\sum (X - \bar{X})^2}$$

4. Intercept (b₀)

$$b_0 = \bar{Y} - b_1 \bar{X}$$

5. Regression Equation

$$\hat{Y} = \text{theta0} + \text{theta1}x$$

6. Residual / Error

$$e = Y - \hat{Y}$$

7. Sum of Squared Errors (SSE)

$$\text{SSE} = \sum (Y - \hat{Y})^2$$

8. Main Objective of Linear Regression

$$\text{Minimize } \sum (Y - \hat{Y})^2$$

Quick Formula Flow

Find $\bar{X}$ and $\bar{Y}$ → Calculate deviations → Calculate slope (b_1) → Calculate intercept (b_0) → Create $\hat{Y} = \text{theta0} + \text{theta1}x$ → Make predictions → Calculate residuals → Minimize SSE

Meaning of Important Symbols

Symbol	Meaning
X	Independent variable / input
Y	Actual dependent variable / target
$\hat{Y}$	Predicted value of Y
$\bar{X}$	Mean of X
$\bar{Y}$	Mean of Y
Theta0	Slope
Theta1	Intercept
n	Number of observations
e	Residual / prediction error
Σ	Summation